



Than Duc Huy

Mobile: (+65) 8642 5472

Email: than_duc_huy@outlook.com

Website (QR):

<https://than-duc-huy.github.io/>

Availability:

- Part-time Internship
- Summer Internship
- Graduate

Jan 2024 – May 2024

May 2024 – Aug 2024

From Jun 2024

EDUCATION

Nanyang Technological University (NTU)

Aug 2020 – Jun 2024 (Expected)

- **Bachelor of Engineering (Electrical and Electrical Engineering)**
- **Honours (Highest Distinction)** (Expected); Current CGPA: 4.92/5.00
- **CN Yang Scholarship Recipient** (Science & Engineering Research)
- Courses: Modelling & Control, Robotics & Automation, Information Security
- Online Courses: Credit-transfer approved by NTU
 - Coursera: Financial Markets (Yale); Simulation and Modelling of Natural Process (University of Geneva)

PROJECTS

NTU-EEE Overseas Final Year Project, Research Exchange at King's College London

Tactile Regrasp of Objects with Dynamic Center-of-Mass

Dec 2022 – May 2023

- Fabricated novel Gelsight vision-based tactile sensor: cast silicone membrane, connected electronics and performed computer vision processing (OpenCV).
- Applied Deep Reinforcement Learning Algorithm: Offline DQN using Gelsight for regrasp tasks.
- Used **ROS**, **Movelit** to implement and control UR5 robot arm and Robotiq-2F gripper.

MLDA@EEE Robotics Team: Team Leader (Team of 4)

Participated in **RoboMaster University Sim2Real Challenge 2023**

Sep 2022 – Jun 2024

- Programmed customized **Python**, **C++** nodes on **ROS** framework, utilized navigation packages (move_base, acml) and behavior_tree to control robot in a simulation environment in **Docker** container.
- Trained other members on ROS framework.

NTU- Design and Innovation Project: **Recipe Management App** (Team of 4)

Aug 2022 – Nov 2022

- Programmed in **Java** to implement OOP Application.
- Incorporated **Github** workflow: raising issues, pushing commits, documenting on Github pages with UML diagrams.

Garage@EEE Project:

6 DOF Stewart Platform with Linear Actuator

Aug 2021 – Jun 2022

- Built encoded linear actuators from standard linear actuators and electrical vernier callipers, keeping cost within budget compared to purchasing commercial encoded actuator.
- Debugged, reverse engineered the vernier callipers with logic analyzer to interpret digital signals.
- Programmed in **C++** 6 microcontrollers for actuators PID control and communication with master microcontroller.
- Applied kinematics for parallel linkage, controlled using bluetooth PS4 controller.

CNYS Making & Tinkering Module: Team Leader (Team of 4)

6 DOF Controller for Robotic Arm Control

May 2021 – Sep 2021

- Built 6 DOF controller with aluminium frame, encoders and robotic arm with Robotis Dynamixel servos.
- Applied forward kinematics concepts, DH parameters to define serial linkage controller; applied inverse kinematics to control servo motors.
- Performed calculation and visualization with **Python**, matplotlib; used Bluetooth communication with ESP32.

CNYS Research Module: **Characterize Time-of-Flight Sensor with Optical Fibre Waveguide**

Jan 2021 – May 2021

- Constructed testing platform using off-the-shelf components and Arduino to create static and dynamic operating conditions.
- Characterized sensor behaviors with optical fiber waveguide to route light to drone exterior.

WORK / INTERSHIPS

Institute for Infocomm Research (I2R), A*STAR

May 2023 – Jul 2023

Research Intern

Sep 2022 – Nov 2022

- Improved laboratory capability by transferring Gelsight fabrication knowledge from KCL.
- Applied Deep Reinforcement Learning algorithms: TD3, PPO using Gelsight for regrasp tasks.
- Explored State Estimation techniques such as Kalman Filter, Factor Graph (gtsam).

Hwa Chong Institution

Teacher Assistant

Jan 2020 – Jul 2020

- Set up IT technology and services to assist teachers in response to COVID home-based and hybrid learning.
- Prepared tutorial and practice worksheets for A Level Physics lessons.

OVERSEAS EXCHANGE PROGRAM

King's College London, United Kingdom (KCL)

Dec 2022 – May 2023

- Research Exchange for Final Year Project
- Gained insight into full-time research experience: proposed and discussed ideas with supervisors and other researchers, developing competency and independence in theorizing and implementation.
- Adapted to social norms and appreciated various cultural values.

LEADERSHIP / CO-CURRICULAR ACTIVITIES

Machine Learning and Data Analytics (MLDA@EEE)

Vice President of Academics (2023-2024)

Member of Academics Committee

Aug 2021 – Jun 2024

- Coordinated with Academics committee members to organize 15 workshops, instructing more than 150 NTU students on ML concepts such as CNN, RNN, Transformer.
- Communicated with MLDA Executive Committee and NTU Management staff to propose initiatives with long-term impact (e.g., ML demonstrations for public display and engagement).
- Provided technical support on ML, computer hardware for other portfolios (Projects, Operations, Marketing).

Garage@EEE Makerspace

Head of Operations (2022-2023)

Member of Operation Committee

Aug 2020 – Jun 2024

- Managed inventory and provided technical support for projects using Makerspace resources.
- Conducted internal training for interested members on machine usage: 3D Printer, Laser Cutter, PCB Miller.
- Repaired, upgraded machines (3D Printers) and used Arduino, Raspberry Pi to create custom automation (e.g., periodic heating of PLA filaments to prevent moisture).

SKILLS / INTERESTS

- **Languages:** English, Vietnamese
- **Project Management:** Microsoft and Google Office Suite, Budget & Timeline Management, Presentation & Communication, Showmanship during Product Demonstration
- **Hardware Prototyping:**
 - Design: Fusion 360, SolidWorks, Altium Designer
 - Fabrication: 3D Printing, Laser Cutting, Oscilloscope, Logic Analyzer
- **Hardware Platform:** Windows, Linux, Arduino, ESP32, NVIDIA Jetson, Raspberry Pi
- **Programming:** Python, C Language, C++, Java, Docker, MATLAB, PlatformIO (for Arduino, ESP32)
- **Machine Learning & Reinforcement Learning:** Pytorch, OpenAI gym
- **Robotics:** ROS & Core Packages, State Estimation, Factor Graph, SLAM, Structure From Motion (SfM)
- **Visual Design:** Adobe Photoshop, Illustrator, Premiere Pro, Procreate, Stable Diffusion (automatic1111 Web UI)